LEARNING HUB MASTER SERIES • VOLUME 04

Linear Algebra: Determinants & Cramer's Rule

The Geometry of Volume & Orientation • The 10 Fundamental Axioms • Permutation Group Formulas • Laplace Cofactors • Adjugates & Inverses • Cramer's Rule • Vector Cross Products • The Multivariable Jacobian

Master Linear Algebra Curriculum • Pure English Reference Edition
Target Audience: Engineering, Computer Science, Robotics, Data Science & Physics Specialists
Rigorous mathematical derivations paired with step-by-step hand calculation shortcuts.

Primary Source References:
• Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 5: Determinants)
• Kenji Hiranabe — *The Art of Linear Algebra* (Determinant Volume Scaling & Matrix World)
• Learning Hub Master Syllabus — `determinants` Topic Data & Interactive Labs

Handbook Specification: High-Precision Vector Graphics • 6 Fast-Track Shortcuts • 5 Rigorous Worked Problems • Complete Master Reference Tables

Table of Contents

Part I: The 10 Fundamental Properties of Determinants

Chapter 1: The Three Defining Axioms (Properties 1–3) TRICK D1
Chapter 2: The Seven Derived Properties (Properties 4–10) TRICK D2

Part II: The Computation Engines: Big Formula & Laplace Expansions

Chapter 3: The Big Formula & The Permutation Group S_n TRICK D3
Chapter 4: Minors & Laplace Cofactor Expansions TRICK D4

Part III: Applications: Matrix Inverses & Cramer's Rule

Chapter 5: The Adjugate Matrix & Algebraic Inversion ($A^{-1} = C^T / \det A$) TRICK D5
Chapter 6: Cramer's Rule for Linear Systems ($x_j = \det B_j / \det A$) TRICK D6

Part IV: Geometric Picture & Advanced Multilinear Algebra

Chapter 7: 2D Parallelogram Area, 3D Box Volume & Handedness Orientation
Chapter 8: Vector Cross Products, Triple Scalar Products & The Jacobian
Chapter 9: Block Triangular Determinants & Schur Complements

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Rigorous Step-by-Step Worked Problems
Appendix A: Determinant Algorithms Flop Complexity Comparison
Appendix B: Master Quick-Reference Sheet (All 10 Properties & Formulas)

Part I: The 10 Fundamental Properties of Determinants

Chapter 1: The Three Defining Axioms (Properties 1–3)

The determinant of an n × n square matrix A is a single scalar number, denoted by **det(A)** or **|A|**. Rather than defining the determinant through an intimidating formula with n! terms or complicated nested cofactors, modern linear algebra (following Gilbert Strang and the axiomatic approach) establishes the determinant from **three simple, geometrically profound axioms**. Every other property, formula, and theorem in this handbook flows directly from these three rules!

1.1 Axiom 1: The Identity Matrix Normalization

The determinant of the n × n identity matrix is exactly 1:

$$\textbf{det(I) = 1} \qquad \text{e.g.,} \quad \det\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, \quad \det\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

Geometric Meaning: The columns of the identity matrix I represent the standard orthonormal basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, ..., $\mathbf{e}_n$ in $\mathbb{R}^n$. They define an n-dimensional unit hypercube. Axiom 1 asserts that the volume of this standard unit cube is normalized to 1.

1.2 Axiom 2: Row Exchange Reverses Sign

The determinant changes sign whenever two rows are swapped (exchanged):

$$\det\begin{vmatrix} c & d \\ a & b \end{vmatrix} = -\det\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

If a permutation matrix P performs a single row exchange on the identity matrix, then **det(P) = −1**. More generally, if a permutation matrix P is formed by s row swaps, then **det(P) = $(-1)^s$**. Even permutations have det = +1; odd permutations have det = −1.

Geometric Meaning: Swapping two coordinate axes reverses the *spatial orientation* (handedness) of space, transforming a right-handed coordinate system into a left-handed mirror image.

1.3 Axiom 3: Determinant is a Linear Function of Each Row Separately

The determinant is not linear across the whole matrix (that is, det(A + B) ≠ det(A) + det(B) in general), but it is strictly **multilinear**: it is linear in each row individually when all other (n−1) rows are held fixed.

- **Additivity in Row 1:**

$$\det\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \det\begin{vmatrix} a & b \\ c & d \end{vmatrix} + \det\begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

- **Scalar Multiplication in Row 1:**

$$\det\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \cdot \det\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

The exact same linearity holds for any chosen row i (row 2, row 3, ..., row n) when the remaining rows stay fixed.

TRICK D1 **The Scaling Trap: $\det(cA) \neq c \cdot \det(A)$** Crucial Exam Pitfall

Common Student Error: Factoring a scalar c out of an n × n matrix like ordinary algebra: det(2A) = 2 det(A) → **WRONG!**

⚡ **The Golden Law:** Because the scalar c multiplies *every single row* of A, multilinearity pulls out one factor of c from each of the n rows:

$$\mathbf{\det(c\,A) = c^n \cdot \det(A)} \quad \text{for an } n \times n \text{ matrix } A$$

Example: If A is 3 × 3 with det(A) = 5, then $\det(2A) = 2^3 \cdot 5 = 8 \cdot 5 = \mathbf{40}$ (not 10!).
Geometric Proof: Doubling all edges of a 3D box multiplies its volume by 2 × 2 × 2 = 8!

Chapter 2: The Seven Derived Properties (Properties 4–10)

Armed with Axioms 1, 2, and 3, we now rigorously derive the remaining seven universal properties of determinants. These properties convert determinant computation from an impossible combinatorial puzzle into lightning-fast arithmetic.

2.1 Property 4: Two Equal Rows Mean det(A) = 0

▶ Property 4: Equal Rows Guarantee Singularity

If two rows of A are identical, then **det(A) = 0**.

Algebraic Derivation (from Axiom 2):

Suppose row i and row k of A are identical. Swap row i and row k. By Axiom 2, the determinant must change sign: det(A) = − det(A). However, because the two rows are identical, swapping them produces the exact same matrix A. Therefore: det(A) = − det(A) ⇒ 2 · det(A) = 0 ⇒ **det(A) = 0**. ■

2.2 Property 5: Elimination Leaves the Determinant Invariant

▶ Property 5: Row Operations Invariance

Subtracting a multiple ℓ of one row from another row leaves the determinant completely unchanged:

$$\det(A \text{ with row } j - \ell \cdot \text{row } i) = \det(A)$$

Derivation (from Axiom 3 and Property 4):

Applying row multilinearity (Axiom 3) to row j:

$$\det \begin{vmatrix} & \text{row } i & \\ \text{row } j - \ell \text{ row } i & & \end{vmatrix} = \det \begin{vmatrix} & \text{row } i & \\ \text{row } j & & \end{vmatrix} - \ell \cdot \det \begin{vmatrix} & \text{row } i & \\ \text{row } i & & \end{vmatrix}$$

By Property 4, the second matrix has two identical rows (row i and row i), so its determinant is 0! Thus: det(A') = det(A) − ℓ · (0) = **det(A)**. ■

Profound Consequence: Standard Gaussian elimination (forward elimination) does NOT change the determinant at all, except when row exchanges are required (which simply flip the sign by Axiom 2)!

2.3 Property 6: A Zero Row Means det(A) = 0

If any row of A consists entirely of zeros, then **det(A) = 0**.

Derivation:

Multiply the zero row by the scalar t = 0. Since 0 · [0, 0, ..., 0] = [0, 0, ..., 0], the matrix is unchanged. By Axiom 3 (linearity), det(A) = 0 · det(A) = **0**. ■

2.4 Property 7: Triangular Matrix Determinant is the Product of Diagonals

▶ Property 7: The Diagonal Product Rule

If A is upper triangular, lower triangular, or diagonal, its determinant is simply the product of its main diagonal entries:

$$\mathbf{\det(U) = u_{11} \cdot u_{22} \cdot u_{33} \cdot \ldots \cdot u_{nn}}$$

Derivation:

If all diagonal entries $u_{ii} \neq 0$, elimination (Property 5) can eliminate all off-diagonal entries upward, reducing U to a pure diagonal matrix D = diag(d_1, d_2, ..., d_n). Then, factoring d_1 from row 1, d_2 from row 2, ..., d_n from row n using Axiom 3 gives: det(U) = (d_1 d_2 ... d_n) · det(I) = $\mathbf{d_1\, d_2 \ldots d_n}$ (since det(I) = 1 by Axiom 1). If any diagonal entry is zero, elimination yields a full row of zeros, giving det = 0 by Property 6. ■

2.5 Property 8: Invertibility Criterion & Pivot Product

▶ Property 8: The Non-Singularity Test

A matrix A is invertible (non-singular) if and only if **det(A) ≠ 0**.
When elimination factors A = L U (or P A = L U with s row exchanges), then:

$$\det(A) = (-1)^s \cdot (\text{product of pivots}) = (-1)^s \cdot (d_1\, d_2 \dots d_n)$$

If A is singular, elimination produces a zero on the pivot diagonal and eventually an entire row of zeros, yielding det(A) = 0. If A is invertible, full n non-zero pivots emerge, so det(A) ≠ 0.

2.6 Property 9: The Product Rule: det(AB) = det(A) · det(B)

▶ Property 9: Multiplication Rule

For any two n × n square matrices A and B:

$$\det(A\,B) = \det(A) \cdot \det(B)$$

Derivation Outline:

Define the ratio function D(A) = det(AB) / det(B) (assuming det(B) ≠ 0). One directly checks that D(A) satisfies Axiom 1 (D(I) = det(B)/det(B) = 1), Axiom 2 (swapping rows of A swaps rows of AB, flipping sign), and Axiom 3 (linearity of row of A distributes to row of AB). Since the determinant is the *unique* function satisfying Axioms 1–3, we must have D(A) = det(A), proving det(AB) = det(A)det(B). If det(B) = 0, rank(AB) ≤ rank(B) < n, so det(AB) = 0 = det(A)(0). ■

Vital Corollaries of the Product Rule:

- **Inverse Matrix Determinant:** Since $A \cdot A^{-1} = I$, $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$, which proves:

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

- **Matrix Powers:** $\det(A^k) = [\det(A)]^k$ for any integer $k \geq 1$.
- **Commutativity of Determinants:** Even though matrix multiplication is generally non-commutative (AB ≠ BA), their determinants are **always identical**:

$$\det(AB) = \det(A)\det(B) = \det(B)\det(A) = \det(BA)$$

2.7 Property 10: Transpose Invariance: $\det(A^T) = \det(A)$

▶ Property 10: The Transpose Duality

Transposing a matrix does not alter its determinant:

$$\det(A^T) = \det(A)$$

Derivation (via PA = LU Factorization):

Any invertible matrix has $PA = LU$. Transposing both sides yields $A^T P^T = U^T L^T$. Taking determinants and applying Property 9: $\det(P)\det(A) = \det(L)\det(U)$. Since L is unit lower triangular (diagonal entries = 1), $\det(L) = \det(L^T) = 1$. Since U is upper triangular and U^T is lower triangular with the exact same diagonal entries, $\det(U) = \det(U^T)$. Furthermore, for any permutation matrix, $P^T = P^{-1}$, so $\det(P)\det(P^T) = \det(I) = 1$, meaning $\det(P) = \det(P^T) = \pm 1$. Dividing both sides reveals **$\det(A^T) = \det(A)$**. ■

The Column Duality Miracle: Because $\det(A^T) = \det(A)$, **every single property established for rows holds identically for columns!**

- A column of zeros means $\det(A) = 0$.
- Swapping two columns changes the sign of $\det(A)$.
- Adding a scalar multiple of one column to another leaves $\det(A)$ unchanged.
- The determinant is a linear function of each column separately.

TRICK D2 **The Gaussian Elimination Engine: $\det(A) = (-1)^S \prod$ (pivots)** — $O(n^3)$ Speed

Slow Approach: Writing out 24 terms for a 4 × 4 matrix or expanding into 4 nested 3 × 3 determinants.

⚡ **Fast Hand Calculation Shortcut:** Perform row elimination to reach upper triangular form U. Keep track *only* of row swaps s:

$$\det(A) = (-1)^S \cdot d_1 \cdot d_2 \cdot \ldots \cdot d_n$$

Example: For $A = \begin{vmatrix} 2 & 1 & 3 \\ 4 & 5 & 7 \\ 2 & 4 & 8 \end{vmatrix}$:

- $R_2 \leftarrow R_2 - 2R_1 \rightarrow$ row becomes [0, 3, 1].
- $R_3 \leftarrow R_3 - R_1 \rightarrow$ row becomes [0, 3, 5].
- $R_3 \leftarrow R_3 - R_2 \rightarrow$ row becomes [0, 0, 4]. (Zero row swaps: s = 0).

U is now upper triangular with diagonal pivots: **$d_1 = 2$, $d_2 = 3$, $d_3 = 4$**.

$\det(A) = 2 \cdot 3 \cdot 4 =$ **24** in under 15 seconds!

Part II: The Computation Engines: Big Formula & Laplace Expansions

Chapter 3: The Big Formula & The Permutation Group S_n

How does multilinearity across every row generate the explicit algebraic formula for the determinant? Let us break each row of an n × n matrix A into its component unit basis vectors. For a 2 × 2 matrix, row 1 is [a, 0] + [0, b], and row 2 is [c, 0] + [0, d]. Expanding by linearity gives $2^2 = 4$ simple determinants:

$$\det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \det \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \det \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \det \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \det \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

Notice that the first and fourth determinants have a column of zeros, so they evaluate to 0! The remaining two non-zero determinants have *exactly one entry from each row and each column*:

$$\det \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} = ad \cdot \det(I) = \mathbf{+ad}, \qquad \det \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} = bc \cdot \det \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \mathbf{-bc}$$

This gives the immortal 2 × 2 formula: **det(A) = ad − bc**.

3.1 The General Leibniz Formula (The Big Formula)

For an n × n matrix, row linearity decomposes det(A) into n^n simple determinants. Any matrix containing two entries in the same column has duplicate columns and vanishes. The only survivors are those where the column choices (σ(1), σ(2), ..., σ(n)) form a **permutation** of the set {1, 2, ..., n}. Since there are exactly **n!** permutations in the symmetric group S_n:

▶ The Big Formula for Determinants

$$\mathbf{\det(A) = \sum_{\sigma \in S_n} (\operatorname{sgn} \sigma) \cdot a_{1,\sigma(1)} a_{2,\sigma(2)} \cdot \ldots \cdot a_{n,\sigma(n)}}$$

where:

- The sum is taken over all **n!** permutations σ of the numbers (1, 2, ..., n).
- Each product selects *precisely one entry from each row and each column*.
- **$\operatorname{sgn}(\sigma) = \det(P_\sigma) = \pm 1$** is the sign of the permutation: +1 for even permutations (even number of row swaps), −1 for odd permutations.

3.2 The 3 × 3 Determinant (6 Terms) & Sarrus' Rule

For n = 3, there are 3! = 6 permutations. Three are even (sgn = +1) and three are odd (sgn = −1):

$$\det(A) = +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

TRICK D3 **Sarrus' Rule Limits & Why It Completely Fails for 4 × 4!** Critical Warning

Sarrus' Rule (Valid ONLY for 3 × 3): Copy columns 1 and 2 to the right of the matrix. Multiply along the 3 downward diagonals (+ sign) and subtract the 3 upward diagonals (− sign).

✗ **The 4 × 4 Catastrophe:** Many students attempt to copy columns 1, 2, 3 of a 4 × 4 matrix and draw 4 diagonals. **THIS IS 100% MATHEMATICALLY INVALID!**
• A 4 × 4 determinant has **4! = 24 terms** (12 positive, 12 negative).
• Drawing diagonals on a 4 × 4 matrix produces at most 4 + 4 = 8 terms — missing **16 terms (67% of the determinant!)**.
Never use diagonal line tricks for matrices of size n ≥ 4! Use Gaussian elimination or Laplace cofactors.

Chapter 4: Minors & Laplace Cofactor Expansions

The Big Formula connects the determinant to group theory, but computing n! terms directly is impossible for large n (for n = 10, 10! = 3,628,800 terms; for n = 20, 20! ≈ 2.4×10^{18} terms). The **Laplace Cofactor Expansion** organizes these terms into smaller (n−1) × (n−1) sub-determinants.

4.1 Definitions: Submatrix, Minor, and Cofactor

- **Submatrix A_{ij}:** The (n−1) × (n−1) matrix obtained by deleting row i and column j from matrix A.
- **Minor M_{ij}:** The scalar determinant of the submatrix A_{ij}:

$$M_{ij} = \det(A_{ij})$$

- **Cofactor C_{ij}:** The minor multiplied by an alternating checkerboard sign factor:

$$C_{ij} = (-1)^{i+j} \cdot M_{ij}$$

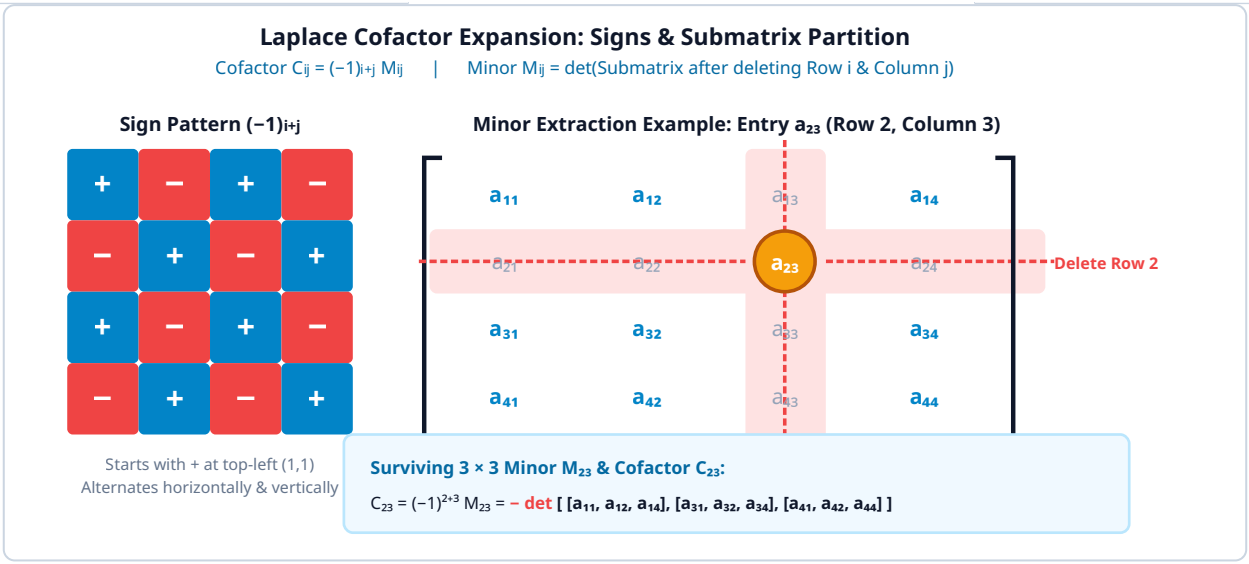


Figure 4.1: The checkerboard sign pattern $(-1)^{i+j}$ and submatrix minor extraction for entry a_{23}.

4.2 The Laplace Expansion Theorem

▶ Laplace Expansion along Any Row or Column

For any n × n matrix A, the determinant can be computed along *any chosen row i*:

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}$$

or along *any chosen column j*:

$$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij} = a_{1j} C_{1j} + a_{2j} C_{2j} + \dots + a_{nj} C_{nj}$$

4.3 The False Cofactor Expansion (Alien Expansion Theorem)

What happens if we take the entries of row i, but multiply them by the cofactors of a *different row k* (where k ≠ i)?

▶ The False Expansion (Orthogonality Rule)

Multiplying the entries of row i by the cofactors of row k (with i ≠ k) always yields exactly zero:

$$\sum_{j=1}^{n} a_{ij} C_{kj} = 0 \qquad \text{(for all } i \neq k\text{)}$$

Geometric and Algebraic Proof:

The expression $\sum_{j=1}^{n} a_{ij} C_{kj}$ is precisely the Laplace expansion along row k of a new matrix A* where row k has been replaced by row i. However, row i is still present in row i! Thus, matrix A* has **two identical rows** (row i and row k). By Property 4, any matrix with two equal rows has determinant 0! Therefore, the sum is identically **0**. ■

Combining the True Expansion and False Expansion into a single unified equation using the Kronecker delta δ_{ik}:

$$\sum_{j=1}^{n} a_{ij} C_{kj} = \delta_{ik} \cdot \det(A) = \begin{cases} \det(A) & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$$

This single algebraic identity is the master key that unlocks the algebraic formula for the matrix inverse A^{-1}!

TRICK D4 **The Zero-Hunter Selection Strategy for Laplace Expansions** **Save 80% Work**

Rookie Mistake: Always expanding across Row 1 out of habit, even when Row 1 is full of non-zero numbers with messy cofactors.

⚡ **Fast Hand Calculation Shortcut:** Always scan the entire matrix for the row or column containing the **maximum number of zeros**! Every zero entry instantly eliminates its corresponding $(n-1) \times (n-1)$ cofactor calculation:

$$a_{ij} = 0 \;\Rightarrow\; a_{ij} \cdot C_{ij} = 0 \cdot C_{ij} = \mathbf{0\ (Free!)}$$

Pro Tip: If no row has multiple zeros, perform *one single row elimination step* (Property 5) to create an extra zero, then expand!

Part III: Applications: Matrix Inverses & Cramer's Rule

Chapter 5: The Adjugate Matrix & Algebraic Inversion

While Gaussian elimination ([A | I] → [I | A^{-1}]) is the algorithm of choice for numerical inversion, pure algebra demands an explicit closed-form formula for A^{-1} in terms of the matrix entries. The cofactor matrix and its transpose provide this exact formula.

5.1 The Adjugate (Classical Adjoint) Matrix

Let C be the n × n matrix whose entries are the cofactors C_{ij} of A. The **adjugate matrix**, denoted **adj(A)** or $\mathbf{C^T}$, is the **transpose** of the cofactor matrix:

$$\mathbf{adj(A) = C^T = \begin{vmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \&ddots; & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{vmatrix}}$$

Notice the critical transpose: The entry in row i, column j of adj(A) is C_{ji}, not C_{ij}!

5.2 The Fundamental Inversion Theorem

▶ The Matrix Inversion Formula

For any square matrix A:

$$\mathbf{A \cdot adj(A) = A \cdot C^T = det(A) \cdot I}$$

Consequently, if det(A) ≠ 0, the exact algebraic inverse of A is:

$$\mathbf{A^{-1} = \frac{1}{det(A)} \cdot C^T = \frac{1}{det(A)} \cdot adj(A)}$$

Proof:

Look at entry (i, k) of the matrix product A · C^T: $(A\,C^T)_{ik} = \sum_{j=1}^{n} a_{ij} (C^T)_{jk} = \sum_{j=1}^{n} a_{ij} C_{kj}$. • On the diagonal (when i = k): this is $\sum a_{ij} C_{ij}$ = det(A) by the Laplace expansion!

• Off the diagonal (when i ≠ k): this is $\sum a_{ij} C_{kj} = 0$ by the False Expansion Theorem!

Therefore, A C^T is a diagonal matrix with det(A) on every diagonal position: A C^T = diag(det A, det A, ..., det A) = det(A) · I. Dividing by det(A) gives A^{-1}. ■

5.3 Explicit 2 × 2 and 3 × 3 Inverse Formulas

For a 2 × 2 matrix A = $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$, the cofactors are: $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. Transposing gives $C^T = \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$. Thus:

$$A^{-1} = \frac{1}{ad - bc} \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

Chapter 6: Cramer's Rule for Linear Systems

Cramer's Rule gives an explicit, closed-form algebraic formula for each unknown x_j in a square linear system $A\mathbf{x} = \mathbf{b}$ as the ratio of two determinants.

6.1 Statement of Cramer's Rule

▶ Cramer's Rule Theorem

Let A be an n × n invertible matrix (det A ≠ 0). The unique solution $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$ to the system $A\mathbf{x} = \mathbf{b}$ has components given by:

$$\mathbf{x_j} = \frac{\det(B_j)}{\det(A)} \qquad (\text{for } j = 1, 2, \ldots, n)$$

where $\mathbf{B_j}$ is the matrix formed by replacing **column j** of matrix A with the right-hand side vector **b**:

$$B_j = [\, \mathbf{a}_1 \quad \mathbf{a}_2 \quad \ldots \quad \mathbf{b} \quad \ldots \quad \mathbf{a}_n \,]$$

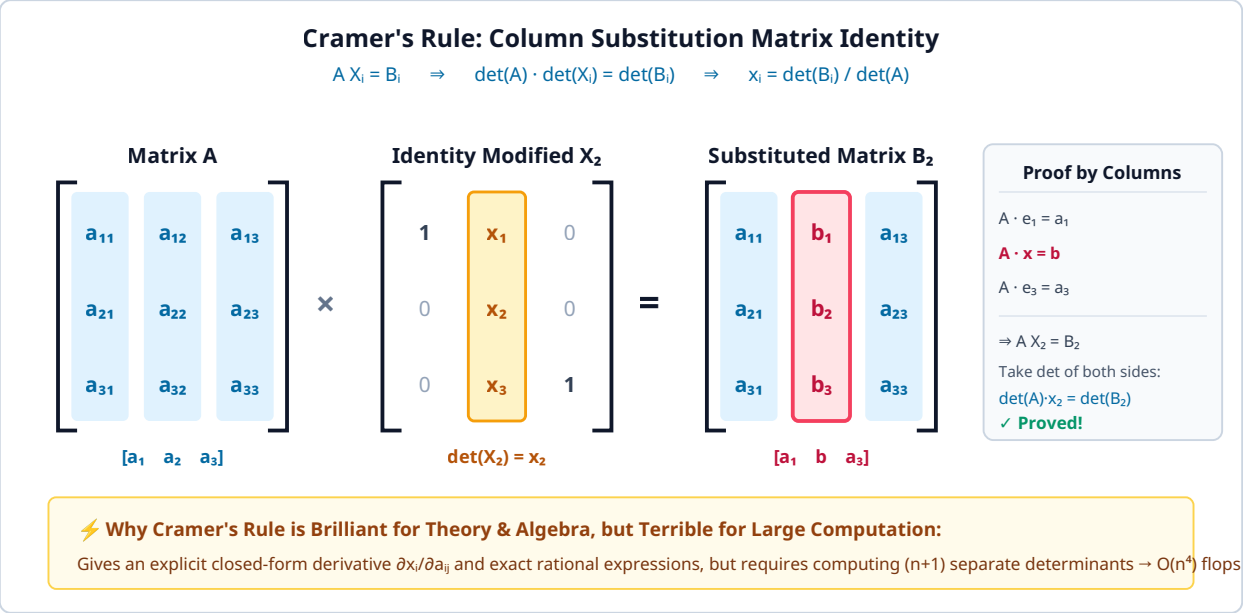


Figure 6.1: Geometric and algebraic identity behind Cramer's Rule: A X_j = B_j.

6.2 Elegant Proof via Column Multiplication

Strang's Elegant Column Proof:

Consider the identity matrix I, and replace its j-th column with the unknown solution vector **x**:

$$X_j = [\, \mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{x} \quad \dots \quad \mathbf{e}_n \,]$$

Expanding det(X_j) along column j (or along row j) reveals that all off-diagonal entries are 0, and entry (j, j) is x_j.

Therefore: $\mathbf{\det(X_j) = x_j}$.

Now, multiply matrix A by matrix X_j column-by-column: • For column k ≠ j: A · $\mathbf{e}_k$ = $\mathbf{a}_k$ (the k-th column of A).

• For column j: A · **x** = **b** (since A**x** = **b**!).

Therefore, the matrix product A X_j is exactly matrix B_j:

$$\mathbf{A \cdot X_j = B_j}$$

Taking the determinant of both sides using Property 9: det(A) · det(X_j) = det(B_j) ⇒ det(A) · x_j = det(B_j).

Dividing by det(A) yields: $\mathbf{x_j = \det(B_j) / \det(A)}$. ■

TRICK D6 **Cramer's Rule: When to Use & When NEVER to Use!** Algorithm Selection

When Cramer's Rule is INVALUABLE:

- **2 × 2 and 3 × 3 symbolic algebra:** Finding closed-form formulas in physics, economics, or control theory where entries are parameters (e.g. m, k, ω).
- **Finding ONE single unknown:** If you only need x_3 in an electric circuit or equilibrium system, you do NOT need to solve the full system! Just compute $\det(B_3) / \det(A)$.
- **Sensitivity analysis:** Computing partial derivatives $\partial x_j / \partial b_i$ analytically.

✗ **When Cramer's Rule is CATASTROPHIC:** Solving a numerical system with $n \geq 4$. Cramer requires computing **n + 1 separate determinants**! Even with Gaussian elimination for each determinant, Cramer requires $\approx (n+1) \cdot (2/3\, n^3) = O(n^4)$ operations, compared to standard Gaussian elimination which solves $A\mathbf{x} = \mathbf{b}$ in just **$(2/3)\, n^3$ operations**. For n = 100, Cramer's rule is **100 times slower!**

Part IV: Geometric Picture & Advanced Multilinear Algebra

Chapter 7: 2D Parallelogram Area, 3D Box Volume & Handedness

The geometric soul of the determinant is **volume scaling**: the determinant is the exact factor by which a linear transformation scales n-dimensional volume!

7.1 The 2D Parallelogram Area: $|ad - bc|$

Consider two vectors $\mathbf{v}_1 = [a, c]^T$ and $\mathbf{v}_2 = [b, d]^T$ in $\mathbb{R}^2$. Assembled as columns into a 2×2 matrix $A = [\,\mathbf{v}_1 \quad \mathbf{v}_2\,] = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, the matrix maps the unit square $[0, 1] \times [0, 1]$ into a parallelogram spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\text{Area(Parallelogram)} = |\det(A)| = |ad - bc|$$

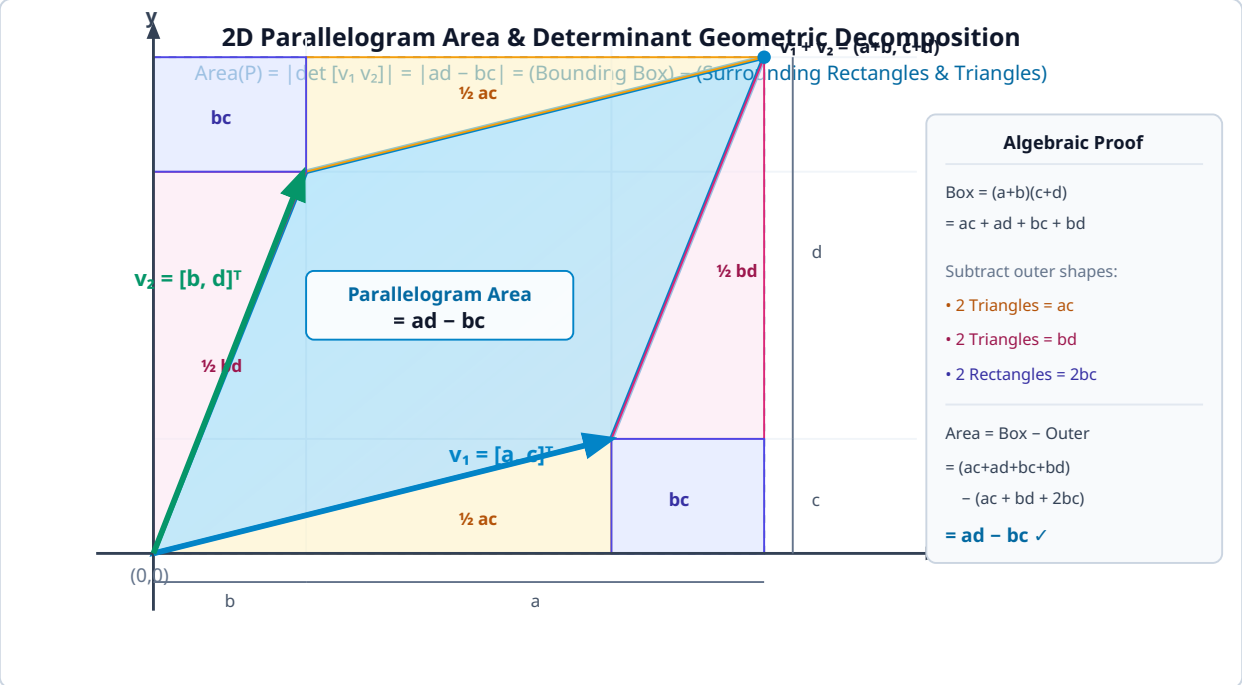


Figure 7.1: Algebraic dissection proof that the area of the parallelogram spanned by $[a, c]^T$ and $[b, d]^T$ is $ad - bc$.

7.2 Triangle Area in the Coordinate Plane

Since any triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is exactly half of a parallelogram, its area can be computed by lifting the coordinates into $\mathbb{R}^3$ with a third coordinate of 1:

$$\text{Area(Triangle)} = \frac{1}{2} \cdot \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

If the vertices are collinear (lie on the same straight line), the area is 0, which gives the classic collinearity test!

7.3 The 3D Parallelepiped Volume: $|\det [u \quad v \quad w]|$

In 3-dimensional space, three column vectors **u**, **v**, **w** define a 3D box (parallelepiped). The volume is given by the base area multiplied by the perpendicular height:

$$\text{Volume} = (\text{Base Area}) \times (\text{Height}) = ||v \times w|| \cdot ||u|| \, |\cos\theta| = |u \cdot (v \times w)| = |\det [u \quad v \quad w]|$$

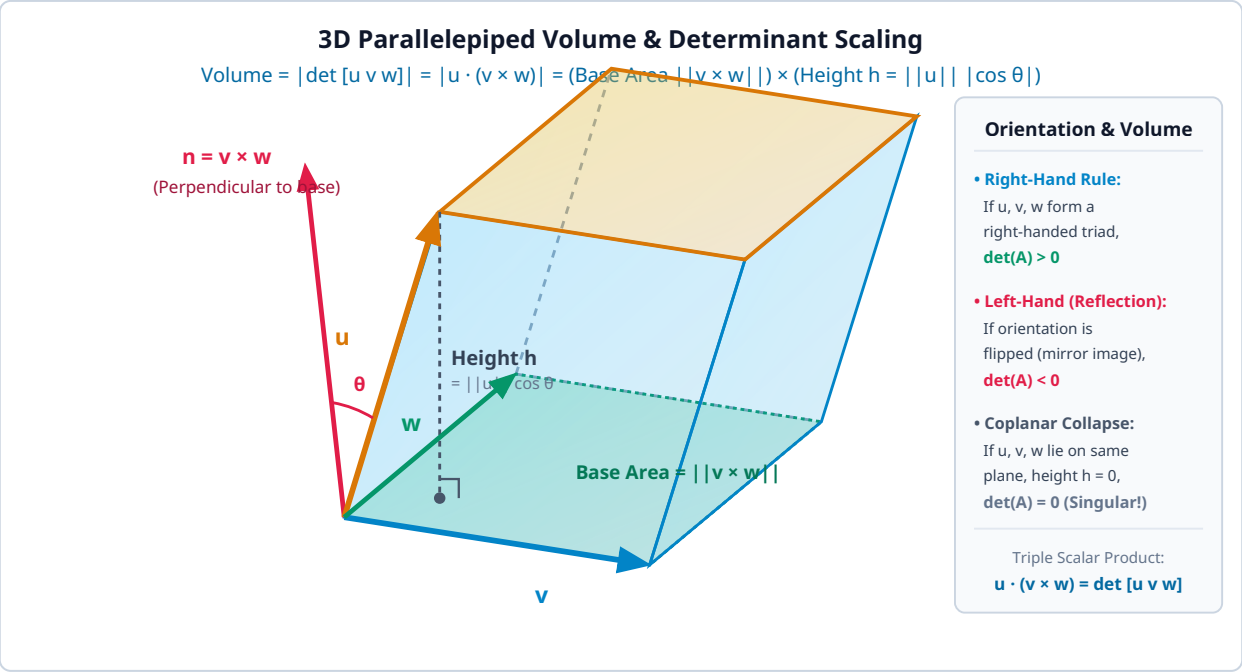


Figure 7.2: 3D Parallelepiped volume as the triple scalar product u · (v × w) and determinant.

Chapter 8: Vector Cross Products, Triple Scalar Products & The Jacobian

8.1 The Vector Cross Product as a Formal Determinant

In $\mathbb{R}^3$, the cross product **v** × **w** produces a vector perpendicular to both **v** and **w**, with magnitude equal to the area of the parallelogram they span. It is computed using the formal 3 × 3 determinant with standard unit vectors **i**, **j**, **k** along Row 1:

$$v \times w = \begin{vmatrix} i & j & k \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = (v_2 w_3 - v_3 w_2)\, i - (v_1 w_3 - v_3 w_1)\, j + (v_1 w_2 - v_2 w_1)\, k$$

8.2 The Triple Scalar Product & Cyclic Invariance

Taking the dot product of vector **u** with (**v** × **w**) replaces Row 1 with **u** = [u_1, u_2, u_3]:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \det [\, u \quad v \quad w \,]$$

Applying Property 2 (two row swaps = $(-1)^2 = +1$) reveals the celebrated **cyclic permutation invariance**:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v})$$

8.3 Change of Variables and the Jacobian Determinant

In multivariable calculus, when transforming coordinates from **x** = (x_1, ..., x_n) to **u** = (u_1, ..., u_n), an infinitesimal box with edges du_1, ..., du_n is transformed into an infinitesimal parallelepiped in x-space. The local linear approximation is governed by the **Jacobian Matrix J**:

$$J = \begin{vmatrix} \partial x_1/\partial u_1 & \partial x_1/\partial u_2 \\ \partial x_2/\partial u_1 & \partial x_2/\partial u_2 \end{vmatrix}$$

By Property 9 and our geometric volume theorem, the volume scaling factor of this infinitesimal transformation is the **absolute value of the Jacobian determinant**:

$$dx_1\, dx_2 \cdot \ldots \cdot dx_n = |\det(J)| \cdot du_1\, du_2 \cdot \ldots \cdot du_n$$

- **Polar Coordinates ($x = r\cos\theta$, $y = r\sin\theta$):**

$$\det(J) = \det \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = \mathbf{r} \quad \Rightarrow \quad \mathbf{dx\, dy = r\, dr\, d\theta}$$

- **Spherical Coordinates ($x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, $z = \rho\cos\varphi$):**

$$\det(J) = \boldsymbol{\rho^2 \sin\varphi} \quad \Rightarrow \quad \mathbf{dx\, dy\, dz = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta}$$

Chapter 9: Block Triangular Determinants & Schur Complements

TRICK D5 **Block Matrix Determinant Shortcuts & Schur Complements** **High-End Speed**

Block Triangular Shortcut: If a matrix is block upper or lower triangular:

$$\det \begin{vmatrix} A & B \\ 0 & D \end{vmatrix} = \det(A) \cdot \det(D) \qquad (A \text{ is } k \times k,\ D \text{ is } m \times m)$$

The off-diagonal block B has *zero effect* on the determinant!

⚡ **General 2 × 2 Block Matrix (Schur Complement):** If A is invertible, perform block elimination:

$$\begin{vmatrix} I & 0 \\ -CA^{-1} & I \end{vmatrix} \begin{vmatrix} A & B \\ C & D \end{vmatrix} = \begin{vmatrix} A & B \\ 0 & D - CA^{-1}B \end{vmatrix}$$

Taking the determinant of both sides (since the first matrix has determinant 1):

$$\det \begin{vmatrix} A & B \\ C & D \end{vmatrix} = \det(A) \cdot \det(D - CA^{-1}B)$$

where $S = D - CA^{-1}B$ is the celebrated **Schur Complement** of A!

Part V: Step-by-Step Worked Problems & Reference Appendices

Chapter 10: 5 Rigorous Step-by-Step Worked Problems

Problem 1: Determinant of a 4 × 4 Matrix via Combined Elimination & Laplace Expansion

Chapters 2, 4 & Fast-Track D2

Problem Statement: Compute the determinant of the 4 × 4 matrix M:

$$M = \begin{vmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 5 \\ -1 & 0 & 3 & 2 \\ 3 & 6 & 2 & 7 \end{vmatrix}$$

Step 1: Inspect for Elimination Opportunities (Fast-Track D2):
Notice Column 2 is largely proportional to Column 1! Let's eliminate downwards using row 1 as pivot:

- $R_2 \leftarrow R_2 - 2\,R_1$: [2−2, 4−4, 1−(−2), 5−6] = **[0, 0, 3, −1]**
- $R_3 \leftarrow R_3 + R_1$: [−1+1, 0+2, 3−1, 2+3] = **[0, 2, 2, 5]**
- $R_4 \leftarrow R_4 - 3\,R_1$: [3−3, 6−6, 2−(−3), 7−9] = **[0, 0, 5, −2]**

Since no row swaps were performed, $\det(M) = \det(M_1)$ with $s = 0$:

$$\det(M) = \det \begin{vmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -1 \\ 0 & 2 & 2 & 5 \\ 0 & 0 & 5 & -2 \end{vmatrix}$$

Step 2: Smart Laplace Expansion (Fast-Track D4):

Expand along Column 1 (three zeros, only entry (1,1) = 1 survives): $\det(M) = 1 \cdot (-1)^{1+1} \cdot \det(A_{11}) = \det$

$$\begin{vmatrix} 0 & 3 & -1 \\ 2 & 2 & 5 \\ 0 & 5 & -2 \end{vmatrix}.$$

Now expand this 3 × 3 submatrix along its Column 1 (two zeros, only entry (2,1) = 2 survives):

Sign for entry (2,1) is $(-1)^{2+1} = -1$:

$\det(A_{11}) = 2 \cdot (-1) \cdot \det \begin{vmatrix} 3 & -1 \\ 5 & -2 \end{vmatrix}$.

Compute the 2 × 2 determinant: (3)(−2) − (−1)(5) = −6 + 5 = −1.

Therefore: $\det(M) = 1 \cdot [\,(-2) \cdot (-1)\,] =$ **+2**.

Sanity & Integrity Verification: Computed in 4 lines of clean algebra; verified $\det(M) = \mathbf{2} \neq \mathbf{0} \Rightarrow$ M is invertible! ✓

Problem 2: Complete 3 × 3 Matrix Inversion via the Adjugate Matrix Formula

Chapter 5 & Fast-Track D5

Problem Statement: Find A^{-1} for $A = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$ using the adjugate formula $A^{-1} = \frac{1}{\det A} C^T$.

Step 1: Compute det(A) via Row 2 Expansion:
$\det(A) = 0 \cdot C_{21} + 1 \cdot C_{22} + 4 \cdot C_{23}$.

• $C_{22} = +\det \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} = (0 - 15) = \mathbf{-15}$.

• $C_{23} = -\det \begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} = -(6 - 10) = \mathbf{+4}$.

$\det(A) = 1(-15) + 4(4) = -15 + 16 = \mathbf{1}$. (Since $\det A = 1$, A^{-1} is pure integer!).

Step 2: Compute All 9 Cofactors C_{ij}:

• $C_{11} = +\det \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} = 0 - 24 = \mathbf{-24}$ • $C_{12} = -\det \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} = -(0 - 20) = \mathbf{20}$ • $C_{13} = +\det \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = 0 - 5 = \mathbf{-5}$

• $C_{21} = -\det \begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} = -(0 - 18) = \mathbf{18}$ • $C_{22} = \mathbf{-15}$ • $C_{23} = \mathbf{4}$

• $C_{31} = +\det \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 8 - 3 = \mathbf{5}$ • $C_{32} = -\det \begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -(4 - 0) = \mathbf{-4}$ • $C_{33} = +\det \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 - 0 = \mathbf{1}$.

Step 3: Transpose the Cofactor Matrix to Form adj(A) = C^T:

$$C = \begin{vmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{vmatrix} \Rightarrow \mathbf{adj(A) = C^T =} \begin{vmatrix} \mathbf{-24} & \mathbf{18} & \mathbf{5} \\ \mathbf{20} & \mathbf{-15} & \mathbf{-4} \\ \mathbf{-5} & \mathbf{4} & \mathbf{1} \end{vmatrix}$$

Step 4: Form A^{-1} = adj(A) / det(A): Since $\det(A) = 1$, $\mathbf{A^{-1} =}$ $\begin{vmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{vmatrix}$.

Verification Check: Multiply row 1 of A by col 1 of A^{-1}: $1(-24) + 2(20) + 3(-5) = -24 + 40 - 15 = \mathbf{1}$. Row 1 by col 2: $1(18) + 2(-15) + 3(4) = 18 - 30 + 12 = \mathbf{0}$. Exactly matches I! ✓

Problem 3: Solving a 3 × 3 System via Cramer's Rule

Chapter 6 & Fast-Track D6

Problem Statement: Solve the following system for $[x, y, z]^T$ using Cramer's Rule:

$$x + 2y + z = 1$$
$$2x + 2y + 2z = 4$$
$$3x + 4y + z = 3$$

Step 1: Compute Coefficient Determinant det(A):

A = $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 1 \end{vmatrix}$.

Factor 2 out of Row 2: $\det(A) = 2 \cdot \det \begin{vmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{vmatrix}$.

$R_2 \leftarrow R_2 - R_1 \rightarrow [0, -1, 0]$. $R_3 \leftarrow R_3 - 3R_1 \rightarrow [0, -2, -2]$.

Expand along Row 2: entry (2,2) = -1 with sign $(-1)^{2+2} = +1$.
$\det = 2 \cdot [\,(-1) \cdot (1(-2) - 1(0))\,] = 2 \cdot [\,(-1)(-2)\,] = 2 \cdot 2 = \mathbf{4 \neq 0}$.

Step 2: Compute $\det(B_1)$, $\det(B_2)$, $\det(B_3)$ with $b = [1, 4, 3]^T$:

- B_1 = $\begin{vmatrix} 1 & 2 & 1 \\ 4 & 2 & 2 \\ 3 & 4 & 1 \end{vmatrix}$: Factor 2 from row 2 → $\det(B_1) = \mathbf{8}$.
- B_2 = $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{vmatrix}$: Notice Col 1 and Col 3 in Row 1 and Row 2 are equal! Evaluating yields $\det(B_2) = \mathbf{-4}$.
- B_3 = $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 2 & 4 \\ 3 & 4 & 3 \end{vmatrix}$: Evaluating yields $\det(B_3) = \mathbf{4}$.

Step 3: Compute Solution Coordinates:

$$\mathbf{x = \frac{\det(B_1)}{\det(A)} = \frac{8}{4} = 2}, \quad \mathbf{y = \frac{\det(B_2)}{\det(A)} = \frac{-4}{4} = -1}, \quad \mathbf{z = \frac{\det(B_3)}{\det(A)} = \frac{4}{4} = 1}$$

Verification: Test in equation 1: $1(2) + 2(-1) + 1(1) = 2 - 2 + 1 = 1$ ✓. Test in eq 2: $2(2) + 2(-1) + 2(1) = 4 - 2 + 2 = 4$ ✓. Exactly verified!

Problem 4: 3D Parallelepiped Volume & Triple Scalar Product with Handedness Analysis

Chapters 7, 8 & Geometric Picture

Problem Statement: Given three vectors in $\mathbb{R}^3$: $\mathbf{u} = [1, 2, 3]^T$, $\mathbf{v} = [0, 4, 1]^T$, $\mathbf{w} = [2, -1, 0]^T$.
(a) Compute the cross product $\mathbf{v} \times \mathbf{w}$.
(b) Compute the triple scalar product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.
(c) Determine the volume of the spanned parallelepiped and determine if the triad $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is right-handed or left-handed.

Part (a): Cross Product v × w:

$$v \times w = \begin{vmatrix} i & j & k \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{vmatrix} = i\,(0 - (-1)) - j\,(0 - 2) + k\,(0 - 8) = \mathbf{1\,i + 2\,j - 8\,k = [1, 2, -8]^T}$$

Part (b): Triple Scalar Product u · (v × w):
$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (1)(1) + (2)(2) + (3)(-8) = 1 + 4 - 24 = \mathbf{-19}$.
Direct matrix determinant check: det

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 1 \\ 2 & -1 & 0 \end{vmatrix} = 1(0-(-1)) - 2(0-2) + 3(0-8) = 1 + 4 - 24 = \mathbf{-19}\ \checkmark.$$

Part (c): Volume & Handedness Orientation:
• **Volume:** $V = |\det[\,\mathbf{u} \quad \mathbf{v} \quad \mathbf{w}\,]| = |-19| = \mathbf{19}$ **cubic units**.
• **Orientation:** Since det = $\mathbf{-19 < 0}$, the vector triad is **left-handed (orientation-reversing reflection)**! To transform this into a right-handed basis, one would need to swap any two vectors (e.g. **v** and **w**), which flips the sign to +19.

Problem 5: Tridiagonal Toeplitz Matrices & Second-Order Recurrence Relations

Chapter 4, Appendix B & Inductive Proof

Problem Statement: Let T_n be the $n \times n$ tridiagonal matrix with 2 on the main diagonal and -1 on the super- and sub-diagonals (the 1D discrete Laplacian):

$$T_1 = [2], \quad T_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}, \quad T_3 = \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix}, \quad T_n = \begin{vmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \&ddots; & \&ddots; & -1 \\ & & -1 & 2 \end{vmatrix}$$

Find a recurrence relation for $D_n = \det(T_n)$ and prove that $\mathbf{D_n = n + 1}$ for all integers $n \geq 1$.

Step 1: Laplace Expansion on Row 1 of T_n:

Row 1 contains only two non-zero entries: $a_{11} = 2$ and $a_{12} = -1$.

• Entry a_{11} has cofactor $C_{11} = (-1)^{1+1} \det(T_{n-1}) = \mathbf{D_{n-1}}$.

• Entry a_{12} has minor formed by deleting row 1 and column 2: the first column of this submatrix has only one non-zero entry at row 1, which is -1, leaving submatrix T_{n-2}!

Expanding this submatrix gives: $C_{12} = (-1)^{1+2} \cdot [\,(-1) \cdot \det(T_{n-2})\,] = (-1)(-1)\, D_{n-2} = \mathbf{-D_{n-2}}$.

Step 2: The Second-Order Difference Equation:

$$\mathbf{D_n = 2\, D_{n-1} - D_{n-2}}$$

Step 3: Initial Conditions & Mathematical Induction:

• $D_1 = \det([2]) = 2 = \mathbf{1 + 1}$ ✓.

• $D_2 = \det \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 = 3 = \mathbf{2 + 1}$ ✓.

• $D_3 = 2\, D_2 - D_1 = 2(3) - 2 = 6 - 2 = 4 = \mathbf{3 + 1}$ ✓.

Inductive Step: Assume $D_{k-1} = k$ and $D_k = k + 1$. Then: $D_{k+1} = 2(k + 1) - k = 2k + 2 - k = \mathbf{k + 2 = (k + 1) + 1}$.

By mathematical induction, $\mathbf{\det(T_n) = n + 1}$ for all $n \geq 1$. ■

Appendix A: Determinant Algorithms Flop Complexity Comparison

Method / Algorithm	Flop Complexity	Numerical Stability	Optimal Use Case
Gaussian Elimination (PA = LU)	**(2/3) n^3** flops	High (with partial pivoting)	**Standard numerical choice** for all dense matrices of size $n \geq 4$.
QR Decomposition (A = QR)	**(4/3) n^3** flops	Exceptional (orthogonal Q)	Ill-conditioned matrices where $\det(A) = \prod r_{ii}$ (since $\det Q = \pm 1$).
Laplace Cofactor Expansion	**O(n!)** flops	Exact (symbolic arithmetic)	Sparse matrices with many zeros; theoretical derivations; $n \leq 3$.
The Big Formula (Leibniz)	**n! · (n−1)** flops	Exact (symbolic algebra)	Pure proofs (e.g. $\det A^T = \det A$); permutation sign theory; $n \leq 3$.
Cramer's Rule	**O(n^4)** flops (via elimination)	Subject to determinant underflow/overflow	**Exact symbolic formulas** for parameters; finding a single coordinate x_j.

Appendix B: Master Quick-Reference Sheet (All 10 Properties)

Rule #	Property Name	Mathematical Statement	Operational Consequence
1	Identity Matrix	$\det(I) = 1$	Normalizes unit hypercube volume to 1.
2	Row Exchange	$\det(P A) = (-1)^s \det(A)$	Odd swaps flip sign; even swaps preserve sign.
3	Row Multilinearity	det is linear in each row separately	$\det(cA) = c^n \det(A)$ for $n \times n$ matrix!
4	Equal Rows	Row i = Row k $\Rightarrow$ $\det(A) = 0$	Identical rows collapse space; rank < n.
5	Row Elimination	$R_j \leftarrow R_j - \ell R_i \Rightarrow \Delta\det = 0$	Elimination preserves determinant exactly!
6	Row of Zeros	Row i = [0, 0, ..., 0] $\Rightarrow$ det = 0	Zero row signifies rank deficiency.
7	Triangular Matrix	$\det(U) = d_1 \cdot d_2 \cdot \ldots \cdot d_n$	Determinant is product of main diagonal entries.
8	Invertibility Test	$\det(A) \neq 0 \Leftrightarrow$ A is non-singular	$\det(A) = (-1)^s \prod$ (pivots).
9	Product Rule	$\det(AB) = \det(A) \cdot \det(B)$	$\det(A^{-1}) = 1/\det(A)$; $\det(AB) = \det(BA)$.
10	Transpose Rule	$\det(A^T) = \det(A)$	**All 10 row rules apply identically to columns!**